

**Communicative Efficiency in Adjective Ordering: Plan-Guided Production
between Hierarchical Semantic Composition and Incremental Pragmatic Choice**

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Abstract

Adjective ordering must reconcile stable order expectations with the usefulness of properties in the current scene. Theoretical accounts locate communicative efficiency in two directions: hierarchical semantic composition interprets noun-adjacent modifiers before outer modifiers, whereas surface production makes linearly earlier information available first. We test how these pressures are expressed in explicit order judgements and in the full distribution of produced utterances. Two German slider studies show a replicated context gradient and a small residual size-first preference when colour alone identifies the target. A production study shows that speakers often resolve reference with one adjective, so adjective selection and utterance length can pre-empt an ordering decision. We compare matched Rational Speech Act models over fifteen completed responses. After controlling participant-specific variation in stable order expectations, plan-guided incremental production predicts responses better than global and fully incremental alternatives. Its advantage spans adjective selection, length, adjective set, and initial adjective. Sequential context updating within hierarchical semantic composition is predictively neutral in the fitted data; controlled simulation identifies conditions under which it can generate an order asymmetry. The results support plan-guided incremental production as an account of completed utterance probabilities and leave the time course of planning open.

Keywords: adjective ordering, reference production, Rational Speech Act, incremental production, Bayesian cognitive modelling

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1 Introduction

Adjective ordering provides a window onto how stable order preferences interact with scene-specific informativeness in referential communication. In languages with prenominal modification, robust ordering preferences favour size-first sequences such as *the big blue sticker* over *the blue big sticker* (Scontras, 2023; Sproat & Shih, 1991). Yet the first adjective is also the listener’s first descriptive cue to the target. In a scene where colour alone identifies the intended sticker, *blue* would provide the decisive information sooner, even though a colour-first sequence conflicts with the canonical size-first preference (Fukumura, 2018; Rubio-Fernández, Mollica & Jara-Ettinger, 2021). The order that sounds most natural may therefore differ from the order that guides the listener most efficiently through the scene.

What counts as efficient—and therefore early—depends on the direction in which an adjective string is evaluated. One line of work concerns adjective subjectivity. More subjective adjectives tend to precede less subjective adjectives in prenominal sequences (Scontras, Degen & Goodman, 2017). Under hierarchical semantic composition, the adjective next to the noun combines with the noun before an adjective farther away. In *the big blue sticker*, *blue* therefore combines with *sticker* before the gradable adjective *big* is interpreted. Because colour is typically less subjective than size, this restriction can provide a more stable comparison class for interpreting size and thereby contribute to a size-first surface preference (Franke, Scontras & Simonic, 2019; Scontras, Bar-Sever, Kachakeche, Rosales Jr & Samonte, 2020; Scontras, Degen & Goodman, 2019).

In reference games (Frank & Goodman, 2012; Franke, 2009), communicative efficiency concerns how an utterance guides the listener to the target as its words unfold. Referential usefulness therefore assigns priority through the surface linear sequence. Listeners encounter the first spoken adjective before the rest of the utterance and can use it to begin restricting the set of possible referents (Rubio-Fernandez & Jara-Ettinger, 2020; Sedivy, Tanenhaus,

Chambers & Carlson, 1999; Tanenhaus, Spivey-Knowlton, Eberhard & Sedivy, 1995). When colour distinguishes the target more effectively than size, placing colour first can make the decisive information available sooner (Fukumura, 2018; Rubio-Fernández et al., 2021). The same size–colour string is therefore traversed in two directions. Hierarchical composition reaches colour before size. Surface linear ordering reaches size before colour. Communicative efficiency can favour different surface orders depending on whether it operates through hierarchical semantic composition or surface linear ordering.

Moreover, before a speaker can weigh alternative adjective orders, they first decide which properties to mention and when to stop. In a colour-sufficient scene, *the blue sticker* may complete the utterance, so competition between size–colour and colour–size orders may never arise. In a size-sufficient scene, speakers may still produce *the big blue sticker*. This creates an asymmetry in the redundant use of colour and size adjectives. Previous empirical studies show that colour is included redundantly more often than relative adjectives such as size (Rubio-Fernández, 2016; Tarenskeen, Broersma & Geurts, 2015). Such redundancy can still support efficient identification, especially when modifier meanings are uncertain and gradable, as with size adjectives (Degen, Hawkins, Graf, Kreiss & Goodman, 2020; Rubio-Fernández, 2016). Scene-specific usefulness may therefore affect which adjectives are selected and when production stops, as well as how jointly selected adjectives are ordered.

These choices make the full distribution of one-, two-, and three-adjective utterances the relevant production outcome. Within this response space, adjective order is conditional on adjective selection and utterance length. This broader response space also raises a question about production architecture. A globally normalised speaker compares complete utterances as wholes. A fully incremental speaker evaluates each next adjective against the alternatives available after the words already produced. Between these endpoints, an utterance-level plan may constrain production while local alternatives continue to influence successive word choices. We call this intermediate hypothesis *plan-guided incremental production*. Under this hypothesis, hierarchical semantic composition can guide an advance

utterance plan, while scene-specific usefulness continues to shape word choice as the utterance unfolds. This possibility connects to broader literature on bounded planning, under which speakers can begin speaking before every part of an utterance has been prepared and can continue planning during articulation (Ferreira & Swets, 2002; Konopka & Meyer, 2014). It also shares with good-enough processing the broader idea that language users can rely on representations whose detail is sufficient for the current task (Ferreira, Bailey & Ferraro, 2002; Frances, 2024). The plan-guided speaker translates this hypothesis into a distributional question by estimating how strongly successive adjective choices affect the probability of each completed utterance.

We therefore ask three questions, moving from the empirical patterns to computational accounts of the mechanisms that may generate them. First, how do stable size-first preferences and scene-specific referential usefulness jointly shape explicit order judgements and the production outcomes of adjective selection, utterance length, and conditional order? Second, after stable order expectations are controlled, does the production distribution favour global evaluation of complete utterances, fully incremental adjective choice, or plan-guided incremental production between these endpoints? Third, does sequential context updating within hierarchical semantic composition improve prediction in the present production data, and under which controlled conditions can it generate an order asymmetry?

Different mechanisms can generate the same surface distribution, making their separate contributions difficult to identify. We therefore combine carefully controlled behavioural experiments, computational cognitive modelling, simulations across a broader range of randomly generated contexts, and Bayesian predictive model comparison. The empirical programme separates explicit ordering preferences from the production choices that make an order observable. Study 1a and its direct replication, Study 1b, measure explicit preferences between size-first and colour-first referential expressions across referential contexts using sliders. Study 2 uses direct production to elicit utterances and records

adjective selection, utterance length, and order conditional on producing the same adjective set. Referential context varies the relative usefulness of size and colour within each scene. Moreover, size discriminability is manipulated between participants, making size a more or less reliable visual cue throughout the experiment. Crossing these factors separates the relative usefulness of size in a particular scene from its perceptual reliability throughout the task. The former directly operationalises discriminatory strength. The latter provides a perceptual-reliability diagnostic relevant to the gradable size meaning invoked by subjectivity-based accounts, with lexical subjectivity held constant.

On the modelling side, we formalise these questions within the Rational Speech Act framework (Degen, 2023; Frank & Goodman, 2012; Goodman & Frank, 2016). We extend the qualitative model of Schlotterbeck and Wang (2023) from two fixed adjective orders to the full production distribution. That account combined sequential context updating within hierarchical semantic composition with incremental speaker choice and left their separate predictive contributions unresolved. All production models assign probabilities to the same fifteen ordered one-, two-, and three-adjective responses. They share controls for stable order expectations, visual salience, adjective selection, and utterance length. One comparison asks whether the interpretation of size remains fixed in the original context or is recomputed through sequential context updating within hierarchical semantic composition. A second comparison asks how strongly successive adjective choices contribute to the probability of a completed response. The global and fully incremental speakers define the two endpoints, and the plan-guided speaker estimates the contribution between them.

Approximate leave-one-out prediction evaluates how accurately these models predict the observed production distribution, with uncertainty in the central contrasts assessed across participants (Vehtari, Gelman & Gabry, 2017). The fitted comparison asks which components add predictive accuracy in the present materials. A separate simulation applies sequential context updating within hierarchical semantic composition across a broader range of randomly generated contexts. The simulation asks when that semantic operation is

sufficient to generate an ordering asymmetry.

Empirically, the slider studies reveal a replicated context gradient: size-first preferences are strongest when size identifies the target and weakest when colour does. A small residual size-first preference remains when colour alone identifies the target. The production study shows that speakers often resolve reference with one adjective, so adjective selection and stopping frequently determine whether an order contrast arises. Lower size discriminability primarily increases the production of redundant adjective material, especially in size-sufficient scenes. After accounting for variation across participants in stable order preferences, predictive accuracy is highest at an intermediate point on the range from global to fully incremental evaluation. The best predictions therefore combine complete-utterance evaluation with continued influence from successive adjective choices. This contribution extends across adjective selection, utterance length, adjective set, and initial adjective. This intermediate architecture is consistent with plan-guided incremental production: advance planning constrains the utterance, and local competition continues to shape successive word choices. Sequential context updating within hierarchical semantic composition adds no meaningful predictive accuracy in the fitted production data. Controlled simulations identify conditions under which sequential context updating within hierarchical semantic composition has stronger semantic consequences and generates an order asymmetry.

The remainder of the paper is structured as follows. Section 2 preserves the two directional accounts of communicative efficiency and relates them to adjective selection and completed-response modelling. Section 3 reports the behavioural studies. Sections 4 and 5 introduce the production model and estimate where prediction is strongest along the global-to-incremental range. Section 6 evaluates the fitted and generative roles of sequential context updating within hierarchical semantic composition, and Section 7 considers the implications and limits of the findings.

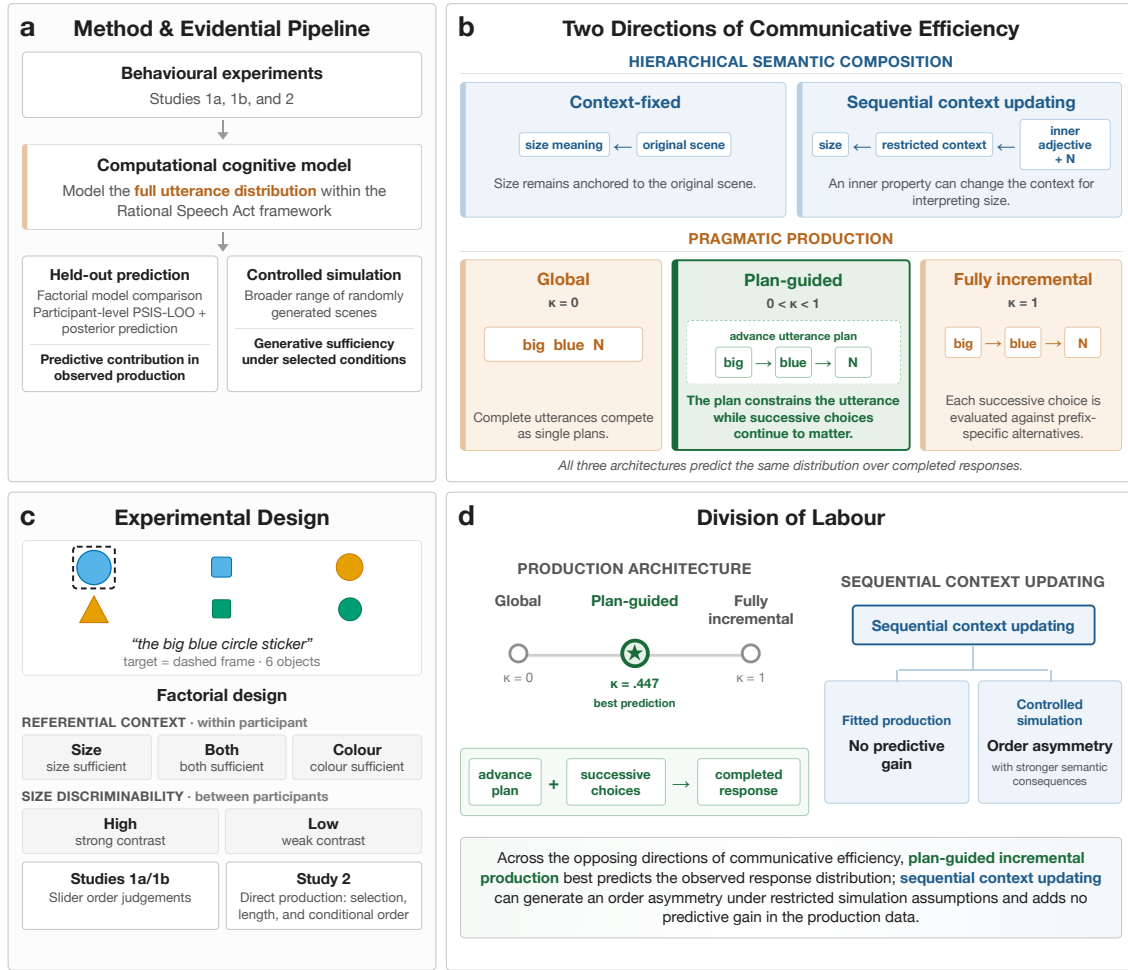


Figure 1

Overview of the design, model space, and evidential structure. Panel A traces the behavioural evidence through computational modelling to held-out prediction and controlled simulation. Panel B distinguishes sequential context updating within hierarchical semantic composition from global, plan-guided, and fully incremental production. Panel C shows the experimental design, and Panel D summarises the predictive result for plan-guided production and the fitted and simulated evidence for sequential context updating.

2 Background

Research on adjective ordering has identified several stable predictors of preferred order, including meaning-based factors such as closeness to the noun and inherentness (Martin, 1969; Sweet, 1898; Whorf, 1945), class-based hierarchies (Cinque, 1994; Dixon, 1982;

Quirk, Greenbaum, Leech & Svartvik, 1972; Scott, 2002), and information-theoretic measures (Dyer, Torres, Scontras & Futrell, 2023; Futrell, Dyer & Scontras, 2020; Hahn, Degen, Goodman, Jurafsky & Futrell, 2018). These predictors explain substantial structure, while residual variation remains even for strong individual predictors (for overviews, see Dyer et al., 2023; Scontras, 2023). Referential context contributes further scene-specific variation in ordering preferences (Fukumura, 2018; Trotzke & Wittenberg, 2019). The present account brings together three lines of work: subjectivity and hierarchical semantic composition, discriminatory strength in reference production, and incremental pragmatic models. Their contributions to the full distribution of one-, two-, and three-adjective utterances remain unresolved.

2.1 Subjectivity, Hierarchical Composition, and Adjective Order

A central empirical generalisation in the adjective-ordering literature is the relation between an adjective’s *subjectivity* and its preferred distance from the noun. Scontras et al. (2017) operationalised subjectivity as susceptibility to faultless disagreement and showed that it explains a large share of variance in English ordering preferences, outperforming earlier predictors such as inherentness and intersectivity. Subsequent work suggests that this relation recurs across languages (Scontras, 2023; Trainin & Shetreet, 2021). It is directly relevant to size–colour sequences because size adjectives are typically more subjective and gradable than colour adjectives, yielding an expected size-first surface preference.

Scontras et al. (2019) locate the explanatory pressure in restrictive composition. Under hierarchical semantic composition, the adjective closest to the noun restricts the nominal denotation before an outer adjective applies. The outer, more subjective adjective is then interpreted relative to a comparison class established by the inner, less subjective adjective. The inner adjective can therefore provide a comparatively stable basis for interpreting the outer gradable adjective. Franke et al. (2019) formalise this intuition in simulation and show that subjectivity-aligned orders can increase average model-assigned target probability under sequential composition with noisy meanings. This account locates

the ordering pressure in hierarchical semantic composition. Scontras et al. (2020) develop the same mechanism as *incremental semantic restriction*. Sequential restriction can therefore generate an ordering pressure when a speaker evaluates complete utterances as wholes.

In the present model, sequential context updating within hierarchical semantic composition implements this proposal. After the inner adjective restricts the relevant context, the interpretation of gradable size can be recomputed relative to that restricted context. Previous simulations establish that this operation can generate an ordering asymmetry under selected semantic assumptions. Its predictive contribution to human production across the full response distribution remains open.

2.2 Referential Usefulness, Adjective Selection, and Length

A second ordering pressure comes from referential usefulness in the surface linear sequence. Fukumura (2018) provide a direct empirical precedent for the present experiments: speakers were more likely to place the more discriminating adjective first. On this account, order reflects which property best distinguishes the target in the current scene. The same adjective can therefore be highly useful in one array and much less useful in another. Rubio-Fernández and colleagues develop this proposal as *incremental communicative efficiency* (Rubio-Fernandez & Jara-Ettinger, 2020; Rubio-Fernández et al., 2021). They argue that an early modifier can be useful because it narrows the listener’s candidate set before the utterance is complete. This proposal accords with evidence that listeners integrate partial linguistic input with visual context (Sedivy et al., 1999; Tanenhaus et al., 1995). From this perspective, word order matters because it determines how the listener’s search space is narrowed as the utterance unfolds.

Referential usefulness also acts before an ordering choice becomes observable. Models of reference planning propose that the contrast set available to the speaker shapes which properties are selected for mention (Fukumura, Van Gompel & Pickering, 2010; Van Gompel, Van Deemter, Gatt, Snoeren & Krahmer, 2019). Perceptual or partner uncertainty can encourage additional, semantically redundant material, and an early redundant modifier can

still facilitate visual search (Degen et al., 2020; Rubio-Fernández, 2016; Rubio-Fernandez, 2019). A one-adjective utterance can resolve reference without creating an order contrast. Slider ratings hold adjective selection and utterance length fixed by presenting two complete orders. They establish an explicit ordering preference and leave open how often production reaches that choice. The relevant production distribution therefore includes one-, two-, and three-adjective utterances.

2.3 Incremental RSA and the Full Production Distribution

The Rational Speech Act framework formalises rational communication in a shared probability space, allowing semantic interpretation, listener inference, and speaker choice to be combined and compared (Degen, 2023; Frank & Goodman, 2012; Goodman & Frank, 2016). Standard RSA models treat complete utterances as the units of speaker choice. The same pragmatic reasoning can be defined incrementally, with the speaker choosing one word at a time relative to the alternatives available after the words already selected (Cohn-Gordon, Goodman & Potts, 2019). Both architectures yield a distribution over completed responses. Variation in planning scope motivates treating global and fully incremental speakers as endpoints for the influence of successive choices (Ferreira & Swets, 2002; Griffin & Bock, 2000; Konopka & Meyer, 2014). Completed responses identify the distributional consequences of these architectures; planning time remains outside the evidence they provide. Related models show how incremental pragmatic reasoning can capture asymmetries in modifier position, order, and redundancy (Degen et al., 2020; Rubio-Fernández et al., 2021).

Schlotterbeck and Wang (2023) introduced the slider paradigm on which Studies 1a and 1b build. Their qualitative RSA model combined sequential context updating within hierarchical semantic composition with incremental speaker choice. Sequential context updating recomputed the interpretation of gradable size as the relevant context was restricted, while the incremental speaker evaluated successive modifier choices. Together, these mechanisms captured a stable size-first preference and its modulation by discriminatory strength. The analysis considered two fixed size-colour orders and left the

separate predictive contributions of the two directions unresolved. The present study extends that starting point with a direct slider replication and a production distribution over one-, two-, and three-adjective utterances, allowing adjective selection, utterance length, and order to be modelled jointly. Within this expanded response space, matched comparisons estimate how much predictive accuracy each direction contributes after stable order expectations and participant variation are controlled. Allowing the contribution of successive adjective choices to vary between the global and fully incremental endpoints provides the distributional test of plan-guided incremental production. A separate simulation asks when sequential context updating within hierarchical semantic composition is sufficient to generate an ordering asymmetry.

3 Experiments

The three studies expose different parts of the same empirical problem. Studies 1a and (HW: its direct replication 1b) ask participants to compare two completed size–colour orders directly. Study 2 instead elicits an utterance and therefore reveals whether speakers mention size, colour, or form, how many adjectives they produce, and only then how selected adjectives are ordered. The tasks can converge in their sensitivity to referential context without measuring the same behavioural construct.

The slider studies test whether a size-first preference is graded by referential usefulness and whether a small residual preference remains when colour alone identifies the referent. The size-discriminability manipulation supplies a secondary display-level diagnostic, but it is not treated as a direct manipulation of lexical subjectivity. The production study tests how context and visual size reliability affect adjective selection, utterance length, initial adjective, and order conditional on producing the same adjective set.

3.1 Shared Method

3.1.1 *Design*

The three studies shared the same basic referential design. The response format differed across experiments. The visual displays and target-definition logic were shared. On

every critical trial, participants saw a six-object display with one framed target. The target was always the largest object that matched the intended colour and form. The slider analyses focus on the theoretically central size–colour subset.¹ Within that subset, referential context varied within participants and had three levels: size-sufficient, colour-sufficient, and both-necessary. A second, between-participant factor, size discriminability, varied the size gap between the target and its size competitors. High size-discriminability displays used larger target–competitor gaps, making the size contrast visually pronounced. Low size-discriminability displays used smaller gaps, making size a less reliable cue. This factor was manipulated between participants so that each participant experienced a consistent level of visual size contrast throughout the task (Figure 2).

3.1.2 *Materials*

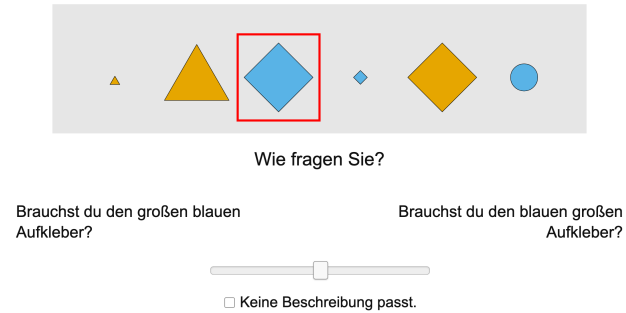
Items were distributed across participants in a Latin-square design, so that scene templates and condition assignments were counterbalanced across lists. In addition to list counterbalancing, the critical scenes were realised through systematic permutations of colour and form values. This created variability in the surface displays and preserved the referential structure required by each condition. These permutations were applied to 27 critical scene templates, each specifying an abstract six-object configuration with a target, competitors, and distractors. Critical vocabularies were fixed across the study, with six colour terms and six form terms reused throughout the scene set; the modified noun was always *Aufkleber* (‘sticker’).² The computational analysis reconstructed each display deterministically from the raw stimulus record as a 6×3 matrix, with one row per object and columns for observed size

¹ Study 2 also included size–form and colour–form trials, which enter the complete production model comparison. The size–colour subset remains the theoretical anchor because it provides the clearest contrast between a gradable adjective and a determinate feature adjective. The appendix reports the exact size values, target–competitor gap construction, and list assignment.

² The form values were presented with German adjectival forms, such as *runden* for round/circle and *quadratischen* for square, so that form could occupy the same prenominal adjective position as colour and size.

A

Design	Levels
dimension	
Referential context	First property sufficient; both necessary; second property sufficient
Size discrimination	High; low
Slider inactivity	
Slider coverage	Size–colour
Production coverage	Size–colour; size–form; colour–form

B**Figure 2**

Experimental design and materials. Panel A summarises the shared design dimensions and task coverage. Panel B reproduces an exact slider-trial screenshot: the framed object is the target, and participants rated the two German adjective orders.

(*s*), colour (*c*), and form (*f*). The referent was stored at index 0. Participants saw the corresponding visual arrays; no stochastic feature recoding enters the final analysis.

3.1.3 Procedure

All studies were conducted online via Prolific. Participants in each experiment were instructed to identify the framed target using its perceptual properties. Spatial descriptions such as *left* or *right* were excluded. The task was framed as a simple reference game about collectible stickers: participants saw one framed target and gave or evaluated a description for a partner who saw the same stickers in a different order and lacked the target frame. This made spatial terms uninformative and kept the task focused on adjective descriptions. Each study began with a consent form and a brief practice block (three training trials with feedback) that familiarised participants with the interface and the referential task. The main

trials were divided into four blocks of approximately equal length, with self-paced breaks between blocks. Trial order within each block was randomised independently for each participant. The experiment-specific response formats, trial counts, and response coding are reported below.

3.2 Study 1a: Slider-Rating Study

3.2.1 *Participants*

Participants ($N = 134$ after exclusions; 70 in the LOW size discriminability group, 64 in the HIGH size discriminability group) were recruited online via Prolific and reported German as their native language. Exclusion criteria were: more than three missing trial-list entries, completion times more than two standard deviations from the sample mean, more than three failures on control items, or more than three “none fits” responses on experimental items.³ After exclusions, the analysed dataset contained 10,606 critical observations. The size-colour subset reported in the main empirical and model-comparison figures contained 3,485 observations; the appendix summarises exclusion details. Mean completion time was 27.6 minutes. Participants were compensated via Prolific at a rate above the platform-recommended minimum.

3.2.2 *Procedure*

Study 1a implemented the shared referential design as a comparative judgment task. On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a “none of the descriptions fits” option. Left-right presentation of the two orders was randomised across trials. Each participant saw one of six lists built from the 27 underlying scenes, yielding 81 critical trials and 99 filler and control trials (180 total). The filler and control items discouraged fixed response strategies and

³ Control items were filler trials with an unambiguous expected response, included to screen for inattentive or non-referential responding. The additional “none fits” option allowed participants to indicate that neither description identified the target; frequent use of this option on experimental items was treated as evidence of a response strategy outside the intended comparative-judgment task.

ensured attention to the referential task. Responses were reoriented so that larger values favoured the size-first order and rescaled from 0–100 to 0–1. For the descriptive figure and linear mixed model, we then subtracted 0.5, yielding a centred scale from -0.5 to 0.5 with zero as the neutral point. The Bayesian predictive analysis retained the same size-first orientation; neutrality corresponds to the midpoint of the response scale.

3.2.3 Results

Figure 3 shows a clear referential-context gradient in size-first ratings. We analysed centred slider ratings with linear mixed-effects models fit in R using `lme4::lmer` (Bates, Mächler, Bolker & Walker, 2014). The reported model included referential context, size discriminability, and their interaction as fixed effects, together with participant intercepts and participant slopes for referential context. Effects were evaluated by likelihood-ratio comparisons between nested models, and planned marginal contrasts were computed with `emmeans`. We complemented these tests with a Bayesian predictive analysis using the same fixed effects and participant-level random-effects structure. The Bayesian analysis evaluates the planned hypotheses through predicted ratings for a new participant and reports posterior medians, 95% credible intervals, and directional posterior probabilities.

Referential context had a significant effect, $\chi^2(2) = 72.2$, $p < .001$. Mean centred ratings, indexing preference for size-first orderings, were highest in size-sufficient displays, lower in both-necessary displays, and lowest in colour-sufficient displays. Mean centred ratings remained above zero in colour-sufficient contexts, yielding no qualitative reversal. Planned contrasts preserved this gradient: size-sufficient ratings exceeded colour-sufficient ratings by 0.268, Bonferroni-corrected 95% CI [0.201, 0.335], and both-necessary ratings exceeded colour-sufficient ratings by 0.172, [0.126, 0.218]. The marginal colour-sufficient estimate was positive at 0.053, and its 95% CI narrowly included zero, $[-0.001, 0.106]$, $p = .056$. The Bayesian predictive analysis assigned posterior probability greater than .999 to the graded context order. It estimated a colour-sufficient preference of .054, 95% credible interval [.003, .109], with posterior probability .982 of exceeding the neutral point.

Size discriminability made a smaller contribution through a significant crossover interaction, $\chi^2(2) = 7.36$, $p = .025$. The planned high–low contrasts within each context included zero, and Study 1b provides the direct test of whether this secondary pattern recurs. The Bayesian predictive discriminability contrasts likewise showed no consistent direction across the three contexts. The central result is therefore a size-first preference graded by referential usefulness: it was strongest when size alone identified the target, weakest when colour alone did, and remained slightly positive in the colour-sufficient condition.

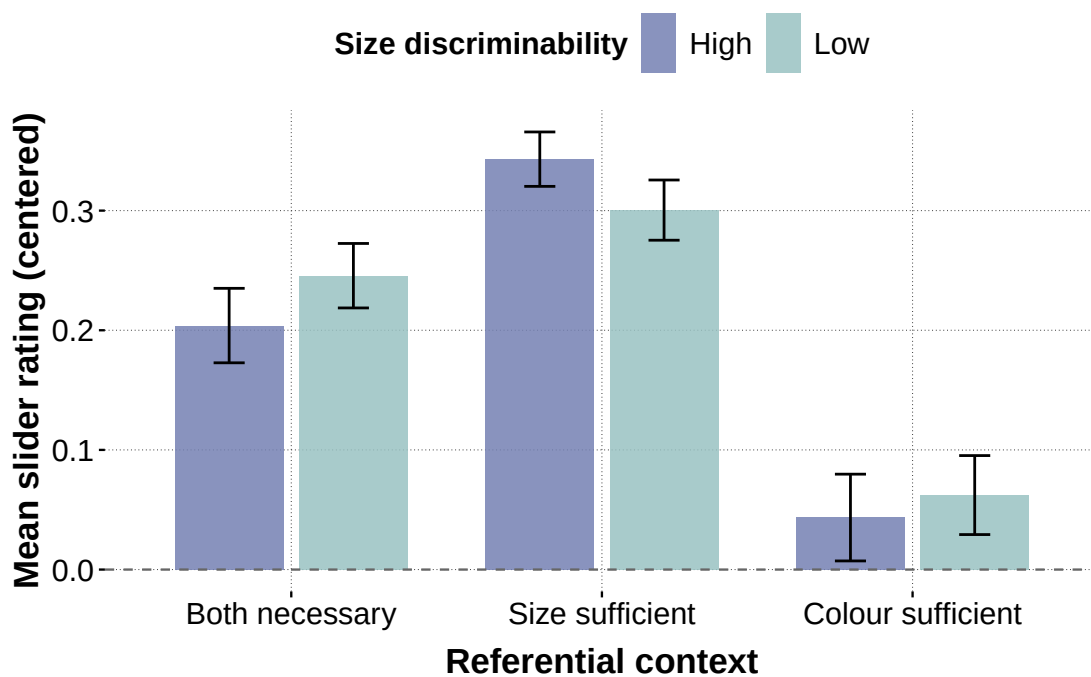


Figure 3

Slider-rating data. Mean centred slider ratings by referential context and size discriminability, with 95% confidence intervals; positive values indicate size-first preference.

3.3 Study 1b: Direct Slider Replication

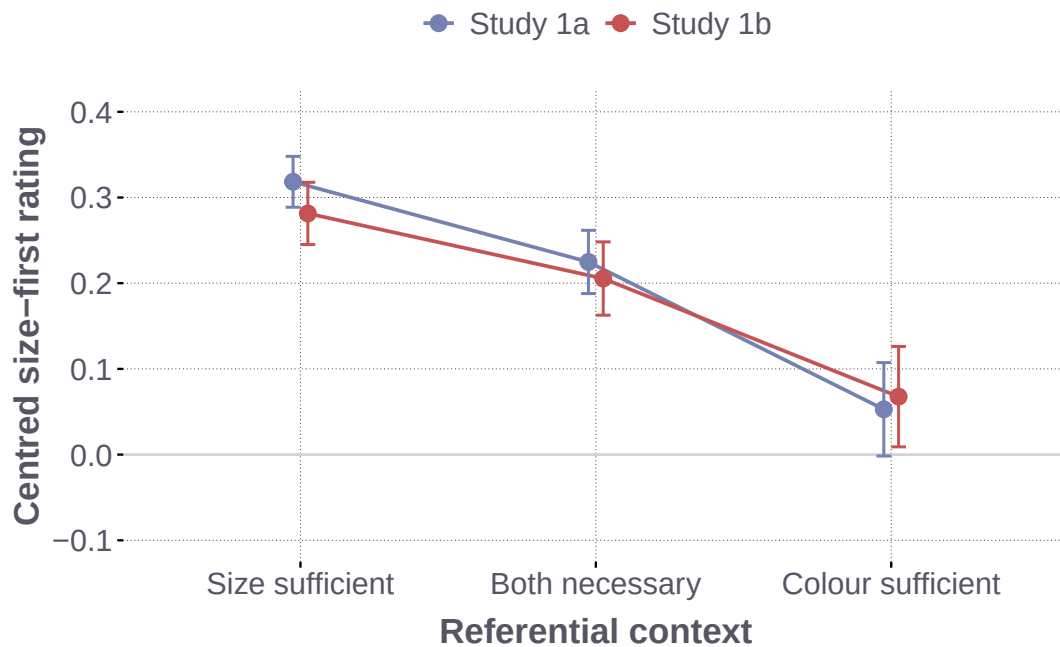
3.3.1 *Participants and preprocessing*

The replication retained 104 of 132 participants after applying the Study 1a exclusions to an independent participant sample. The cleaned file contains the same 1,080 list–item–condition design tuples as Study 1a and 8,379 critical observations. The complete

exclusion and preprocessing record appears in Appendix A.

3.3.2 Results

We fitted the same mixed-effects specification as in Study 1a to the replication ratings. Referential context was again significant, $\chi^2(2) = 38.651$, $p < .001$. Mean centred ratings were 0.272 in size-sufficient contexts, 0.197 when both properties were necessary, and 0.054 in colour-sufficient contexts. The corresponding 95% confidence intervals were [0.235, 0.309], [0.154, 0.240], and [-0.005, 0.113]. All three Bonferroni-corrected context contrasts were significant. The colour-sufficient point estimate virtually reproduced Study 1a (.054 vs. .053), and both confidence intervals narrowly include zero. The matched Bayesian predictive analysis assigned posterior probability .999 to the graded context order. It estimated the colour-sufficient preference at .057, 95% credible interval [.001, .116], with posterior probability .977 of exceeding neutrality. Study 1b therefore directly replicates the primary context gradient and supplies convergent evidence for a small positive colour-sufficient preference. The weak Study 1a crossover did not recur, so size discriminability remains secondary to the replicated context pattern.

**Figure 4**

Combined Study 1a and Study 1b slider results. Mean centred ratings by referential context in each study, with 95% confidence intervals over participant-level means; positive values indicate size-first preference.

3.4 Study 2: Production Study

3.4.1 Participants

Participants ($N = 113$ after exclusions; 60 in the LOW size discriminability group, 53 in the HIGH size discriminability group) were recruited online via Prolific; anyone who had completed either slider study was excluded. Further exclusion criteria were: completion times more than two standard deviations from the mean, more than 80% identical responses, or more than 10% missing or unannotatable responses. The canonical production table contains 9,100 annotated observations across all three adjective-pair families. The model comparison retains the native fifteen-category response space across size-colour, size-form, and colour-form trials. Adjective family is a structured dimension of the materials that specifies which two properties support reference; the primary experimental factors remain

referential context and size discriminability. The targeted frequentist analysis uses size–colour to align directly with the slider studies, while size–form provides generalisation to a second size-containing family and colour–form provides a control for the full-data model comparison. Mean completion time was 32.4 minutes. Compensation followed the same Prolific rate as Study 1a.

3.4.2 Procedure

Study 2 used the shared referential design in a production format. Participants saw the framed target together with the fixed sentence frame *den _____ Aufkleber* and typed the adjective material needed before the noun *Aufkleber*. Restricting the response to the adjective slot reduced off-task variability and provided a controlled basis for coding. The practice materials additionally introduced the relevant colour and form vocabulary before the critical trials began. Each participant completed 135 main trials: 81 critical trials and 54 filler or control trials from one of six counterbalanced lists.⁴

Responses were coded into 15 utterance categories corresponding to all ordered combinations of S (size), C (colour), and F (form): 3 one-adjective, 6 two-adjective, and 6 three-adjective strings. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous; spatial descriptions, comparative or ordinal size expressions, and other unclassifiable responses were excluded. The dependent variable for the Bayesian model comparison is the categorical utterance label $y_i \in \{1, \dots, 15\}$ for each trial i . For supplementary comparison with the slider studies, we also tracked two derived binary projections: whether an utterance was size-initial and whether it contained more adjective material than the minimally sufficient property set. These projections combine several underlying choices and are not treated as direct measures of adjective order. For the mixed-effects logistic regressions, trials satisfying the coded outcome were coded as 1 and all other annotated trials were coded as 0.

⁴ The number of filler and control trials was reduced in Study 2 to keep the total number of trials manageable because typed production responses were slower and more effortful than slider judgments.

3.4.3 *Results*

Figure 5 provides an overview of the ordered response distribution for the 3,034 size–colour trials, which align directly with the slider studies and the targeted behavioural tests. The computational analysis below retains all 9,100 production trials across size–colour, size–form, and colour–form. The complete categorical distribution shows that referential context affects which adjectives are selected and how much adjective material is produced. Both-necessary scenes generally elicit longer descriptions, whereas one-property-sufficient scenes more often elicit a single adjective. In particular, colour-sufficient size–colour scenes are commonly resolved with colour alone. On those trials, no size–colour ordering decision is observed.

Selection, adjective set, and length. In the size–colour trials, speakers preferentially mentioned properties that distinguished the target in the current scene. The resulting unordered adjective-set distribution separates one-property responses from responses that contain both referentially relevant properties and from three-adjective descriptions. Lower size discriminability increased the probability of additional adjective material most clearly when size alone would otherwise have been sufficient. Figure 6A isolates this pattern through the over-informative-response recoding used in the targeted hypothesis test. In the supplementary binary projection, over-informative responses were more frequent under low than high size discriminability in size-sufficient displays (.90 vs. .63). The Bayesian predictive difference was .234, 95% credible interval [.151, .318], and the targeted interaction against the other contexts was .153, [.072, .236]. The visual manipulation therefore bears most clearly on adjective selection and utterance length rather than on a stable ordering preference.

Initial adjective and conditional order. Initial-adjective probabilities reflect both whether an adjective is selected and where it occurs. For this reason, size-initiality is a useful descriptive projection of the complete response space but not a direct counterpart of the slider outcome. Figure 6B shows the size-initial recoding used in the targeted hypothesis

test. Within the size–colour trials, the context-by-discriminability interaction for size-initial responses was significant, $\chi^2(2) = 11.7$, $p = .003$. High discriminability increased size-initial responses in both-necessary displays (.49 vs. .65), with a Bayesian predictive difference of .141, 95% credible interval [.036, .242]. The corresponding effects were small in the two one-property-sufficient contexts. Conditional on producing both size and colour, size-first descriptions remained more frequent than colour-first descriptions; colour-before-size reversals were comparatively rare. That conditional subset is selected by the prior decision to mention both properties and is therefore reported separately from adjective selection and length. The same distinction applies to size–form, colour–form, and three-adjective responses.

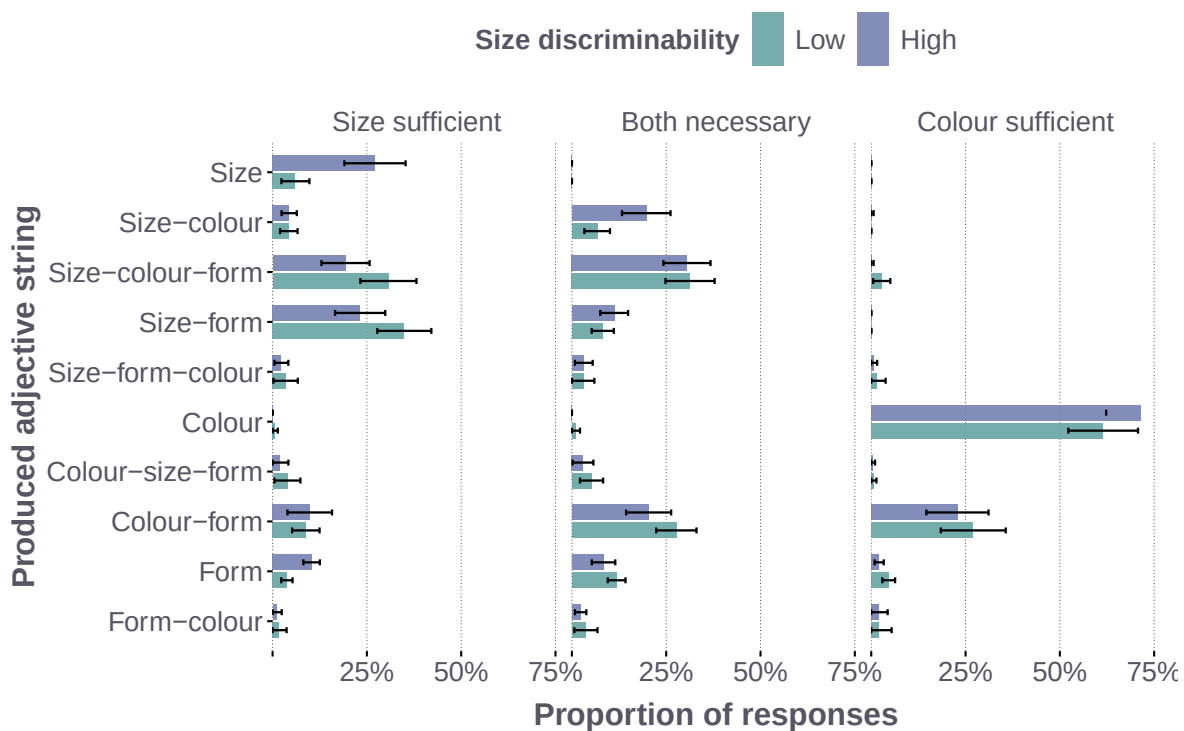
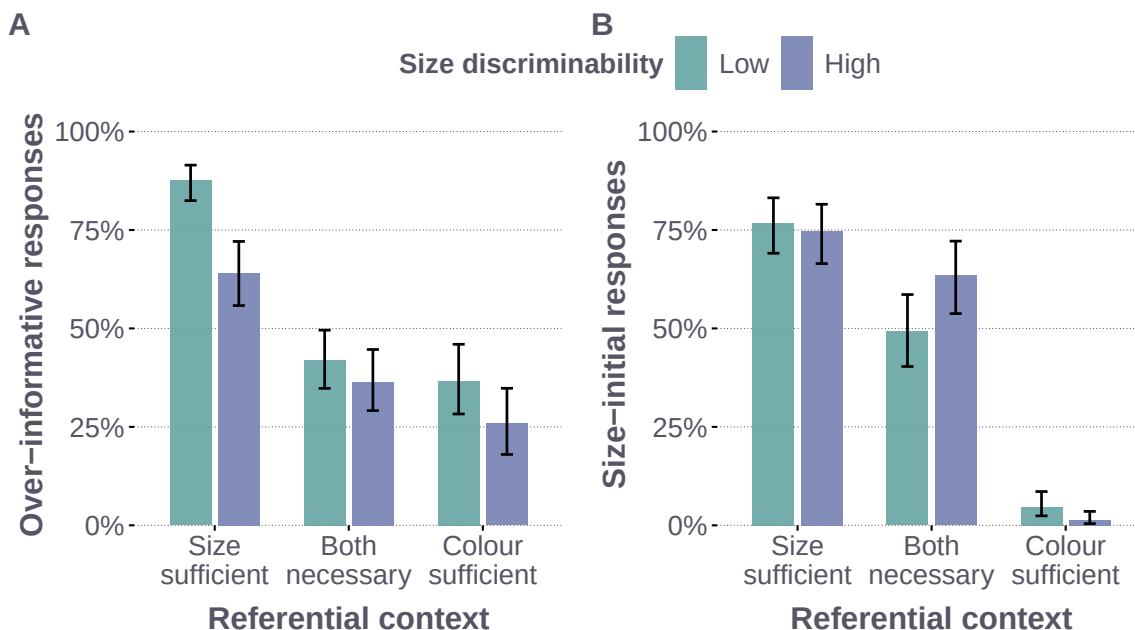


Figure 5

Ordered response distribution across the 3,034 size–colour production trials. Bars show mean participant-level proportions by referential context and size discriminability, with 95% confidence intervals. Adjective labels preserve surface order; the figure shows responses reaching at least 2% in one condition.

**Figure 6**

Hypothesis-linked recodings of the size-colour production responses. Panel A codes redundant adjective use as over-informative: a multi-adjective response in a one-property-sufficient context or a three-adjective response when both properties are necessary. Panel B codes whether the response begins with size. Bars show Bayesian posterior medians and error bars show 95% credible intervals. Size-initiality combines adjective selection and order.

3.5 Behavioural Synthesis

The slider studies establish a replicated context gradient in explicit order preference. Ratings are highest when size identifies the target, lowest when colour does, and retain a small positive size-first estimate in the colour-sufficient condition. The size-discriminability interaction in Study 1a does not replicate and remains secondary.

Production exposes a different decision structure. Scene-specific usefulness affects adjective selection and utterance length, and one-adjective descriptions frequently resolve reference before an order among two adjectives arises. The size-discriminability manipulation primarily changes whether speakers add material, especially in size-sufficient scenes.

Initial-adjective summaries inherit both these selection effects and any subsequent ordering

preference. The formal model must therefore predict the complete fifteen-category response distribution rather than an isolated size-first/colour-first choice.

4 Modelling the Full Production Distribution

The modelling follows the research questions in two steps. The production comparison asks whether speakers evaluate complete utterances globally, choose adjectives fully incrementally, or combine an advance utterance plan with successive adjective choices. The semantic comparison asks whether sequential context updating within hierarchical semantic composition contributes to prediction when a noun-adjacent adjective restricts the comparison class for gradable size. The main text reports these scientifically interpretable comparisons. The Appendix documents the common display representation, stable-order control, task link for adjective selection and utterance length, and supporting robustness analyses.

4.1 Complete Responses and the Literal Listener

The model assigns probabilities to all fifteen ordered strings formed from size, colour, and form: three one-adjective utterances, six two-adjective utterances, and six three-adjective utterances. This response space allows the model to predict which properties speakers mention, how many adjectives they produce, and how jointly selected adjectives are ordered. All models use the observed six-object display and a Rational Speech Act speaker who favours utterances that guide a listener to the target (Degen, 2023; Frank & Goodman, 2012; Goodman & Frank, 2016). Size has a graded meaning defined relative to a comparison class (Franke et al., 2019; Schmidt, Goodman, Barner & Tenenbaum, 2009). Colour and form provide additional visual information about the target.

Let $u_{\leq t} = w_1 \dots w_t$ be a surface prefix and $P_0(o)$ the prior probability of object o in the display. Under context-fixed semantics, the literal listener after this prefix is

$$Q_t(o \mid u_{\leq t}) \propto P_0(o) \prod_{j=1}^t \llbracket w_j \rrbracket_{\text{fix}}(o). \quad (1)$$

The listener combines the meanings made available by the surface prefix and assigns

probability to every object that remains compatible with them. This listener provides the target-directed value used by all three production accounts.

4.2 Successive Adjective Choice and Plan-Guided Production

At position t , the utility of candidate adjective a for participant i and target o^* is

$$r_{it}(a; u_{<t}, o^*) = \alpha_i \log Q_t(o^* \mid u_{<t} + a) + \lambda_{\text{salience}} z_a, \quad (2)$$

where α_i controls sensitivity to listener success and z_a represents the visual salience of the candidate property. For a completed utterance u , the model records both the accumulated utility of its selected adjectives and the competition they faced at successive positions:

$$R_i(u, o^*) = \sum_t r_{it}(w_t; u_{<t}, o^*), \quad (3)$$

$$N_i(u, o^*) = \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp r_{it}(a; u_{<t}, o^*), \quad (4)$$

where $\mathcal{A}(u_{<t})$ contains the adjective types still available after the current prefix. The term R_i scores the selected path through the utterance. The term N_i accumulates the local normalisers associated with the alternatives available along that path. All three accounts therefore use the target-directed values of intermediate prefixes contained in R_i .

The probability of a completed utterance is

$$S_{\kappa,i}(u \mid o^*, X) \propto \exp\{R_i(u, o^*) - \kappa N_i(u, o^*) + \beta_i \Omega(u) + T(u, X)\}, \quad (5)$$

where $\Omega(u)$ is a German language model's relative score for alternative orders of the same adjectives and β_i allows its influence to vary across participants. The task link $T(u, X)$ captures adjective selection and utterance length through the observed effects of adjective sufficiency, form use, size reliability, and length. These terms are shared across the three production accounts.

At common values of the shared parameters, the same model can be written as a geometric interpolation between the global and fully incremental endpoints:

$$S_{\kappa,i}(u \mid o^*, X) \propto S_{G,i}(u \mid o^*, X)^{1-\kappa} S_{I,i}(u \mid o^*, X)^\kappa, \quad (6)$$

where $S_{G,i}$ and $S_{I,i}$ are the endpoint distributions evaluated with the same listener, stable-order component, task link, and participant parameters. The proportionality in Equation (6) renormalises the interpolated scores over the fifteen completed utterances.

The parameter κ determines how strongly competition at successive adjective positions contributes to the completed-utterance probability. With $\kappa = 0$, the local normalisers drop out and the *global* speaker normalises over the fifteen completed paths. With $\kappa = 1$, the *fully incremental* speaker retains the complete contribution of locally normalised adjective choices. An interior value defines the *plan-guided* speaker: an advance utterance plan constrains the completed response, and competition among locally available adjectives continues to shape its surface realisation. Stopping enters through $T(u, X)$ at the completed-response level. Accordingly, κ measures the contribution of successive adjective choice. Online stopping and the time course of planning remain outside its interpretation.

4.3 Sequential Context Updating in Hierarchical Composition

The semantic comparison concerns the inner-to-outer direction of hierarchical composition. Let $c_1, \dots, c_T$ denote the modifiers in a completed utterance ordered from the noun outwards, so that $c_1 = w_T$ for prenominal modification. Starting from $H_0 = P_0$, composition successively restricts the listener state:

$$H_j(o) \propto H_{j-1}(o) \llbracket c_j \rrbracket_{\rho_j}(o), \quad j = 1, \dots, T. \quad (7)$$

For *big blue sticker*, this path incorporates *blue* before the gradable meaning of *big* applies.

The semantic regime determines the comparison class used for size:

$$\llbracket S \rrbracket_{\rho_j}(o) = \begin{cases} \llbracket S \rrbracket_{\theta_\delta(\mathcal{C})}(o), & \text{context-fixed,} \\ \llbracket S \rrbracket_{\theta_\delta(H_{j-1})}(o), & \text{sequential context updating,} \end{cases} \quad (8)$$

where θ_δ derives a graded size threshold from a comparison class and $\mathcal{C}$ is the original scene. Under context-fixed semantics, this threshold remains anchored to the original scene. Under sequential context updating, the threshold is recomputed from the listener state left by the hierarchically prior modifier. The surface-prefix listener in Equation (1) follows the produced

words from left to right. The hierarchical listener H_j follows semantic composition from the noun outwards. This comparison asks whether recomputing the size comparison class adds predictive accuracy in the observed production data. A separate simulation asks when the same operation is sufficient to create an order asymmetry across a broader range of contexts. The threshold construction and further semantic details appear in Appendices B and G.

4.4 Matched Comparisons and Claim Scope

Each final comparison varies one theoretically relevant component and holds the remaining model structure constant. The production comparison evaluates global, plan-guided, and fully incremental production with the same semantics, response space, task link, and participant structure. Stable order expectations vary across participants, allowing the comparison to isolate the additional contribution of successive adjective choices. Participant-specific variation in that contribution and participant-specific variation in stable order expectations are empirically indistinguishable in the present data. We therefore estimate one shared κ and treat participant-level planning profiles as outside the identified claim. The semantic comparison holds the production account fixed and varies whether hierarchical composition uses a fixed or sequentially updated comparison class. Development checks for the task link and other shared model components are reported in the Appendix.

4.5 Inference and Predictive Target

We fit each model to the same 9,100 production trials using Bayesian inference in NumPyro (Phan, Pradhan & Jankowiak, 2019). Approximate leave-one-out prediction compares out-of-sample accuracy for the complete fifteen-response distribution (Vehtari et al., 2017). Because observations are repeated within participants, uncertainty in the central contrasts is additionally aggregated and bootstrapped across participants. The predictive target is a held-out completed response from the sampled task population. Online planning time remains outside this target.

5 Predictive Evidence for Plan-Guided Production

5.1 Population-Level Architecture

The final comparison asks whether the production distribution is better predicted by global, fully incremental, or plan-guided incremental production. All three models use the same listener, response space, stable-order control, task link, and participant structure. The comparison varies the contribution of successive adjective choices to the probability of the completed utterance. The remaining components are held constant.

Plan-guided incremental production exceeded global production by 229.605 ELPD units, with participant-clustered $SE = 81.097$ and a participant-bootstrap 95% interval of $[58.765, 376.160]$. It exceeded fully incremental production by 422.387 ELPD units, with participant-clustered $SE = 108.714$ and a participant-bootstrap 95% interval of $[221.275, 647.382]$. The direct comparison between the two endpoints was unresolved: fully incremental minus global production was -192.782 ELPD, with a participant-bootstrap 95% interval of $[-553.149, 108.336]$. The model comparison therefore favours an account in which completed utterance probabilities reflect both an advance utterance plan and competition among successive adjective choices.

5.2 Contribution of Successive Adjective Choices

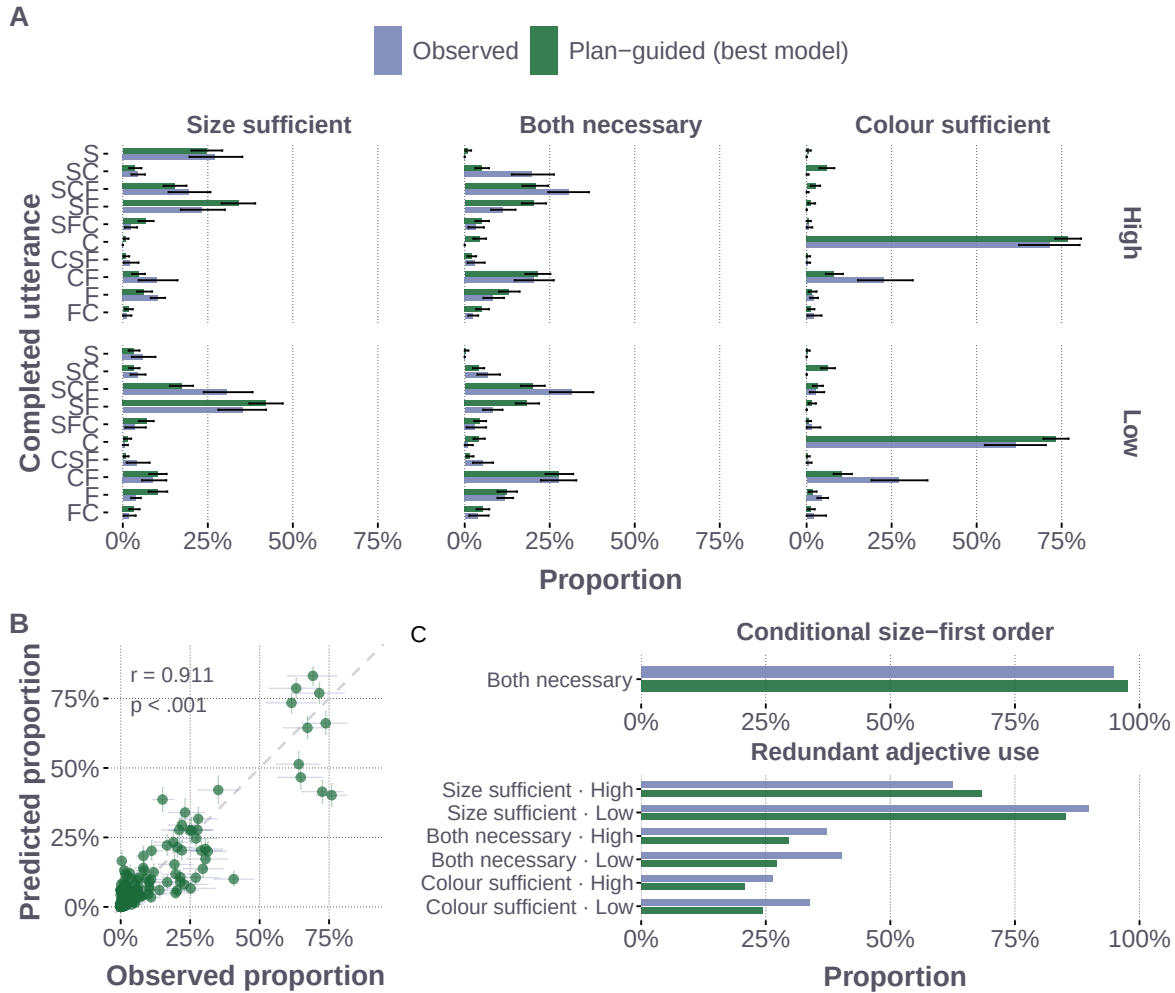
The estimated contribution of successive adjective choices was intermediate between the global and fully incremental endpoints:

$$\kappa = .4472, \quad 95\% \text{ credible interval} = [.4036, .4897]. \quad (9)$$

The posterior lies between the two endpoint accounts under the final specification. The precise value remains conditional on the listener representation, stable-order control, and task link. Its scientific interpretation is therefore qualitative: successive choices contribute to the completed production distribution, and an advance utterance plan continues to constrain that distribution.

5.3 Posterior-Predictive Signature of Plan-Guided Production

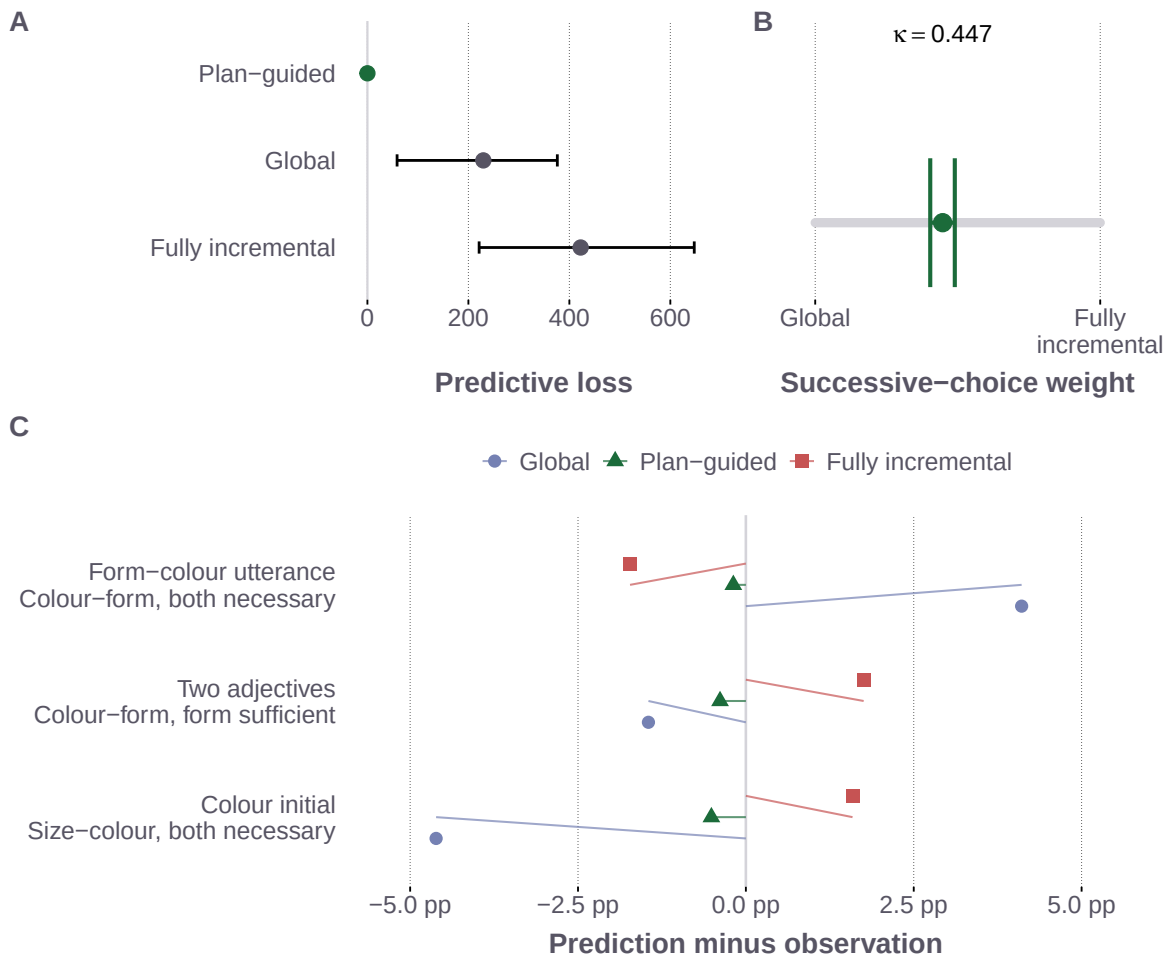
Figure 7 evaluates how well the plan-guided model reproduces the full production distribution. Panel A compares the observed and posterior-predicted proportions of the most frequent completed utterances in the size–colour trials, where the central ordering contrast arises. The model captures the broad shifts between one- and multi-adjective utterances across conditions. Several cells retain visible discrepancies. Panel B extends the check to all fifteen completed utterances across the three adjective families, referential contexts, and size-discriminability conditions, with uncertainty shown for both observed and predicted proportions. The correlation was $r = .911$ ($p < .001$). The p -value is descriptive because cells from the same condition form a compositional distribution and are therefore dependent. Panel C returns to two outcomes from the empirical analysis: size-first order conditional on producing the fixed size–colour adjective set, and redundant adjective use across referential contexts. The posterior-predictive check assesses absolute descriptive adequacy. The leave-one-out comparison evaluates the relative predictive performance of the three architectures.

**Figure 7**

Posterior-predictive adequacy of the plan-guided model. Panel A shows observed and posterior-predicted proportions with their respective uncertainty intervals for frequent size-colour utterances. Panel B compares all 270 cells; horizontal intervals show participant-bootstrap uncertainty in the observations, vertical intervals show 95% posterior-predictive uncertainty, and the dashed line marks equality. Panel C compares observed and posterior-predicted size-first order conditional on the fixed size-colour adjective set in both-necessary trials, collapsed across size discriminability, and redundant adjective use across the referential contexts and size-discriminability conditions used in the empirical analysis. The correlation is descriptive because cells within a condition are dependent.

5.4 Where Plan-Guided Production Improves Prediction

Figure 8 combines the relative predictive comparison with a magnified view of three theory-relevant production outcomes. Panel A shows the leave-one-out predictive loss of the global and fully incremental endpoints relative to plan-guided production, and Panel B locates the posterior estimate of κ between these endpoints. Panel C expresses prediction errors in percentage points for a colour-initial response, a two-adjective response, and a completed form-colour utterance. These outcomes respectively illustrate initial adjective choice, utterance length when one property is sufficient, and the probability of a particular multi-adjective order. The expanded scale makes the small differences among architectures visible while preserving their absolute magnitude. In each example, plan-guided production lies closer to the observed proportion than either endpoint, showing how an intermediate contribution from successive choices can improve several stages of the production distribution.

**Figure 8**

Predictive evidence and residual signature of plan-guided production. Panel A reports leave-one-out predictive loss relative to plan-guided production; intervals are participant-bootstrap 95% intervals. Panel B places the posterior estimate of κ between the global and fully incremental endpoints; the interval is a 95% credible interval. Panel C shows prediction minus observation in percentage points for three production outcomes selected to represent initial adjective choice, utterance length, and a completed multi-adjective utterance. The residual axis is expanded to show the small relative differences among the three architectures.

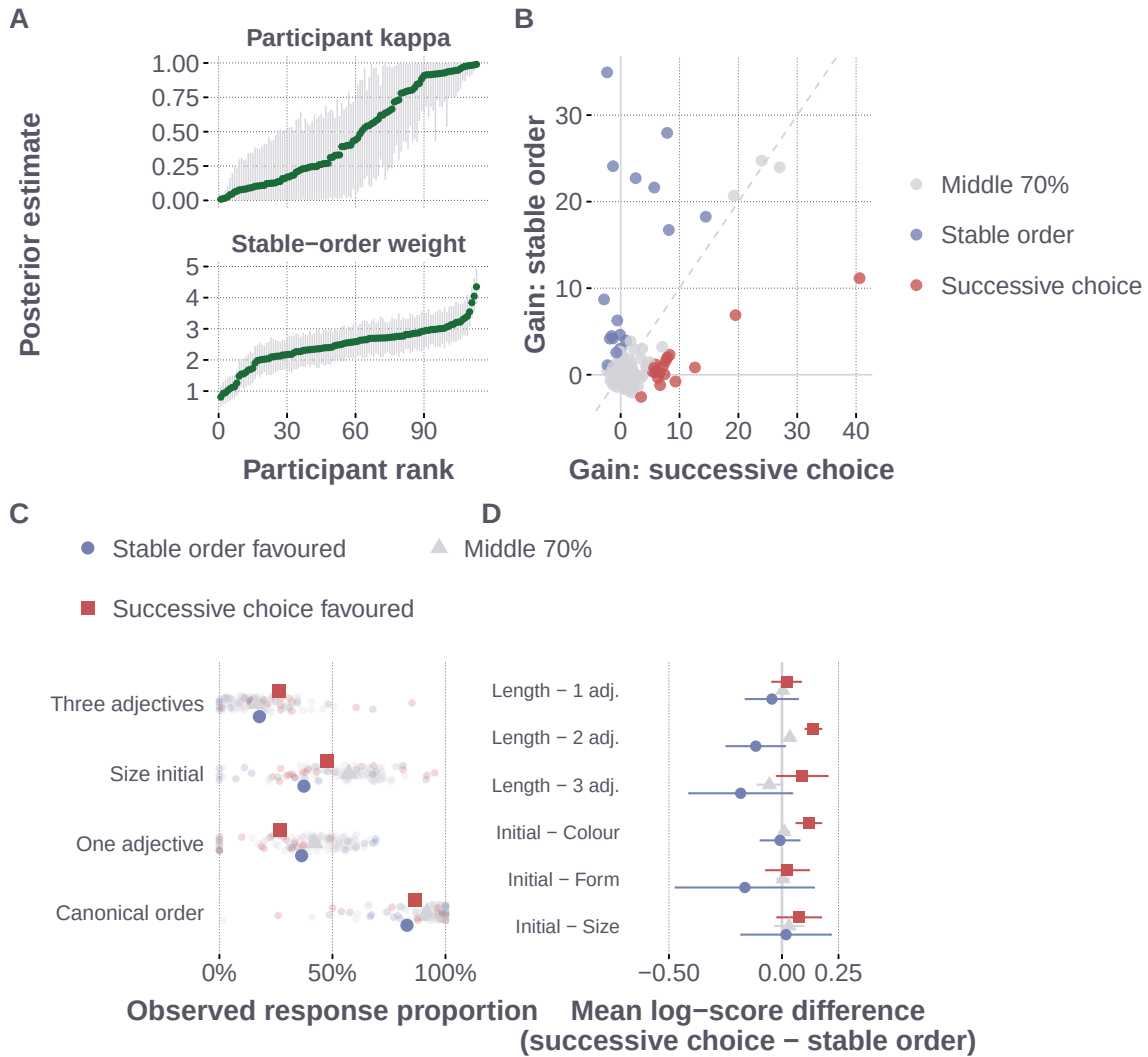
5.5 Locating Participant Variation

The population-level comparison estimates one shared κ while allowing stable order expectations to vary across participants. We examined the location of participant variation through three matched hierarchical extensions. Participant-specific κ_i allows speakers to differ in how strongly successive adjective choices influence the probability of a completed utterance. Participant-specific stable-order weighting allows speakers to differ in their conventional order expectations. Participant-specific length bias allows speakers to differ in their general tendency to continue producing modifiers.

The hierarchical κ_i model and the participant-specific stable-order model yield nearly identical predictive gains over the shared model. Their direct difference, hierarchical κ_i minus participant-specific stable-order weighting, was -2.184 ELPD, with participant-bootstrap interval $[-154.051, 137.077]$. Participant-specific length bias added essentially no predictive accuracy. The hierarchical κ_i fit also missed the registered diagnostic thresholds, with maximum $\hat{R} = 1.0133$ and two observations with Pareto $k > .7$. Completed utterances therefore reveal substantial participant variation. Its source remains unresolved between successive adjective choice and stable order expectations. The final specification uses participant-specific stable-order weighting as the control and estimates one population-level value of κ ; its fit passed the registered diagnostics. The resulting population estimate supports plan-guided production after stable order variation is controlled. Individual planning profiles remain outside the identified claim.

Figure 9 shows where this unresolved participant variation appears in the response distribution. Panel A displays posterior means and 95% credible intervals for participant-specific κ_i and stable-order weights in the two matched hierarchical models. Panel B shows that the two forms of participant variation often improve prediction for the same participants. Participants at the extremes can favour different parameterisations. For an exploratory residual description, Panels C and D divide participants by the upper and lower 15% of their participant-level predictive difference between these models. Panel C

shows the participant observations underlying these response profiles together with the three group means. The group relatively favouring participant-specific successive choice produced fewer one-adjective responses and more three-adjective responses. The group relatively favouring stable-order variation showed a lower rate of size-initial responses in multi-adjective utterances. Panel D localises the corresponding log-predictive differences across utterance length and initial adjective on one signed scale. Positive values indicate higher predictive accuracy for participant-specific successive choice, and negative values indicate higher predictive accuracy for participant-specific stable-order weighting. These groups are defined and described using the same responses, so they identify response patterns associated with the residual difference. Stable participant strategies remain unidentified.

**Figure 9**

Participant variation and its unresolved source. Panel A shows participant-level posterior means and 95% credible intervals, and Panel B compares leave-one-out predictive gains over the shared model. Panel C shows observed participant proportions as pale points and group means as larger symbols for the lower 15%, middle 70%, and upper 15% of the contrast between hierarchical models. Panel D shows the mean leave-one-out log-score difference for the same groups across response categories; horizontal intervals are participant-level 95% confidence intervals. Positive values favour participant-specific successive-choice contribution, and negative values favour participant-specific stable-order weighting. The κ_i fit missed the registered diagnostic thresholds, and the groups provide an exploratory residual description.

5.6 Task Link and Claim Scope

A reduced task link preserved the relative predictive advantage of plan-guided production and substantially reduced absolute predictive accuracy. The production account therefore requires both a model of how utterances are evaluated and a task link for adjective selection and length. The full model-development and participant-identification analyses appear in Appendices D and E.

6 Sequential Context Updating within Hierarchical Semantic Composition

6.1 Predictive Contribution in the Production Data

The third research question asks whether sequential context updating within hierarchical semantic composition contributes to the production distribution. Both models preserve the same inner-to-outer semantic hierarchy. They differ in whether the interpretation of size remains fixed in the original scene or is recomputed after the noun-adjacent adjective restricts the relevant context.

The matched comparison found effectively equal predictive accuracy: updating minus fixed semantics was -0.26 ELPD, with participant-bootstrap interval $[-0.97, 0.44]$. Under the production constants and actual size-colour displays, recomputing the size threshold after a colour restriction changed the target’s graded size value and the literal listener’s target probability by less than .001 on average. The fitted materials therefore place the model in a regime where sequential context updating has extremely small semantic consequences.

The formal inner-to-outer hierarchy is retained in both models. The predictive result concerns the additional proposal that the comparison class for size is dynamically recomputed along that hierarchy. The simulation evaluates the same proposal under conditions in which its semantic consequences are larger.

6.2 When Can Sequential Context Updating Create an Order Asymmetry?

The simulation removes the fitted stable-order component, visual-salience calibration, task link, and participant variation. Both surface orders are interpreted by a hierarchically

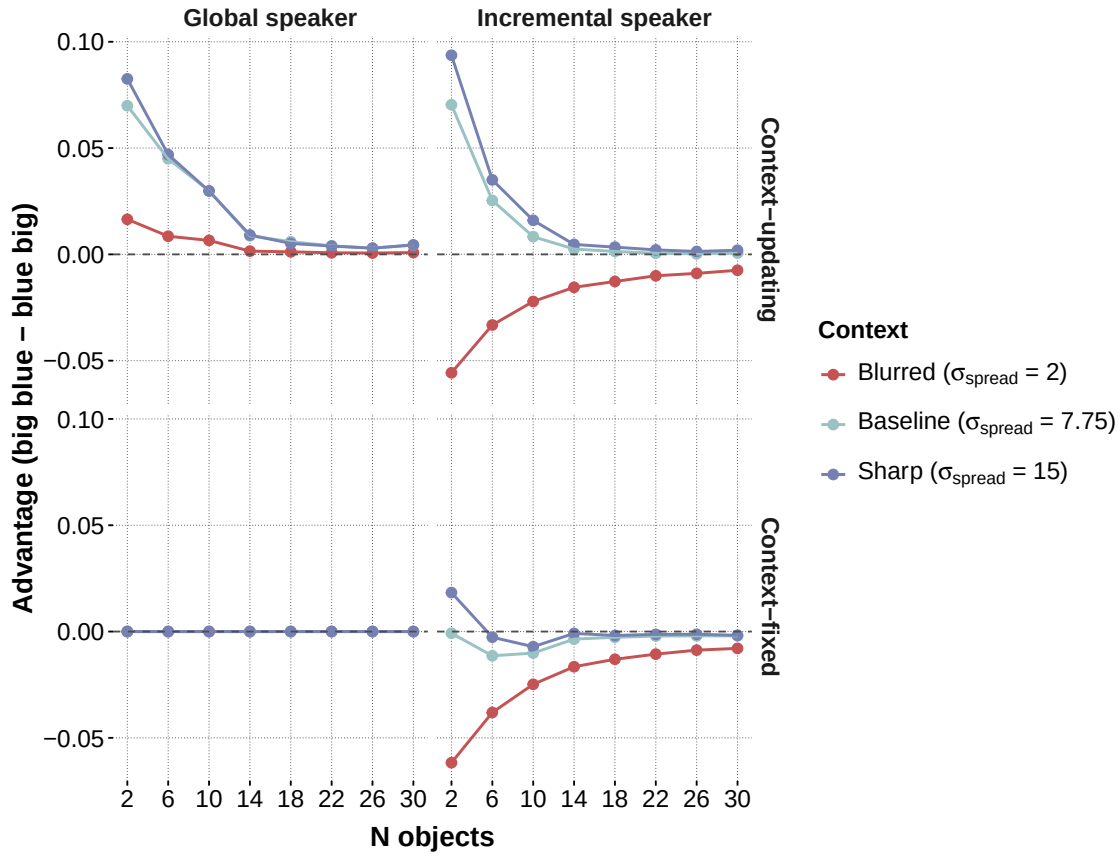
compositional listener that combines the noun-adjacent adjective first. We cross global versus fully incremental production with fixed versus sequentially updated comparison classes over a broad grid of context sizes, size distributions, colour reliability, threshold selectivity, and perceptual blur. Full grid details appear in Appendix H.

For each simulated context, we compute the pragmatic listener’s target probability after *big blue* and after *blue big*. The size-first advantage is

$$\Delta_{\text{SF}} = L_1(o^* \mid \text{“big blue”}) - L_1(o^* \mid \text{“blue big”}). \quad (10)$$

Positive values indicate a higher model-assigned target probability for the size-first order.

Sequential context updating produces a positive size-first advantage across much of the high-reliability simulation grid, including under global production. The fixed-context global model assigns the same target probability to the two orders. Fully incremental production changes the shape of the asymmetry and can yield a colour-first advantage when size is weak and the context is small. The simulation therefore establishes generative sufficiency and boundary conditions. Its evidential target is the behaviour generated by the model under controlled assumptions.

**Figure 10**

Boundary simulation of the size-first advantage Δ_{SF} as a function of context size and contextual spread, faceted by production account and semantic regime. Positive values indicate a higher model-assigned target probability for the size-first order. The figure characterises model behaviour under simulated assumptions and leaves participants' use of the mechanism open.

7 General Discussion

The behavioural studies place adjective ordering within the broader problem of reference generation. Explicit slider judgements exhibit a replicated context gradient: size-first preference is strongest when size identifies the target, weakest when colour does, and remains slightly positive in colour-sufficient scenes. Production distributes the same contextual pressure across a larger response space. Speakers often select one sufficient adjective, and visual size reliability primarily affects whether they add material. On those trials, selection and utterance length determine whether a multi-adjective ordering decision arises at all.

The final production comparison evaluates that complete response space. After participant-specific variation in stable order expectations is controlled, plan-guided incremental production outpredicts both global and fully incremental production. The estimated contribution of successive adjective choices lies between these endpoints, and the predictive gains extend across adjective families, response lengths, adjective sets, and initial adjectives. This is the paper’s central computational result. It concerns the probability distribution over completed utterances and leaves the temporal implementation of planning open.

Stable order expectations enter through a German language-model residual whose influence varies across participants. The model treats this conventional size-first tendency as a controlled source of variation. Accurate prediction of adjective selection and length additionally requires a task link for salience, adjective sufficiency, form use, visual reliability, and utterance length. A reduced task link preserves the relative advantage of plan-guided production and loses substantial absolute predictive accuracy. Plan-guided production therefore operates within a broader account of the task decisions that make an adjective order observable.

The two directional accounts of communicative efficiency receive different forms of evidence. Sequential context updating within hierarchical semantic composition makes no

practically meaningful predictive contribution in the present materials. Controlled simulation shows that the same operation can create a size-first target-probability advantage when colour is more reliable and the semantic update is less diffuse. That result establishes a possible semantic route under the implemented assumptions. Participants' use of that route remains open, leaving an empirically identified division of labour between semantics and production beyond the present evidence.

The participant hierarchy yields a second negative identification result. Participant-specific planning weights and participant-specific stable-order weights perform indistinguishably. The completed responses therefore reveal systematic participant consistency while leaving its source unidentified. The final model retains participant order variation as an auxiliary control and one shared population κ . Individual planning profiles, stable strategies, and cross-session characteristics remain outside the interpretation.

Hierarchical composition specifies which semantic restriction is computed inside another in a formal representation. Plan-guided production specifies how an advance utterance plan and successive surface choices jointly shape completed utterance probabilities. The millisecond time course of semantic interpretation and utterance planning remains open for both operations. Online measures would be needed to identify temporal implementation.

The experiments use German prenominal noun phrases in a controlled reference task with a fixed adjective vocabulary and a restricted fifteen-response support. The treatment of form supports prediction, and the stable order component is imported from a language model. Further work can test whether the architecture generalises to freer production, new lexical families, different adjective systems, and designs that independently manipulate lexical subjectivity and perceptual reliability. Such extensions should preserve the distinction between response-space prediction, semantic sufficiency, and process identification.

The strongest conclusion is therefore bounded and substantive. Adjective order is an outcome of choices about what to mention and how much to say. Within that response-generation problem, completed utterances are predicted best by plan-guided

incremental production, in which an advance utterance plan constrains production while successive adjective choices continue to shape the final response.

8 Data Availability Statement

Data, analysis scripts, model code, final posterior summaries, and figure-generation scripts will be made available in an anonymised repository upon submission. The repository will include the deterministic production input, the final global, plan-guided, and fully incremental model artifacts, preprocessing and reconstruction receipts, and the scripts required to reproduce the reported behavioural and predictive analyses.

9 Author Note

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Appendix A

Methods and Coding Transparency

Table A1 summarises the participant exclusions and the analysis tables used in the reported manuscript analyses. The per-criterion exclusion counts are not mutually exclusive, so the unique number of excluded participants is smaller than the sum of the criterion counts. For Study 2, the canonical production table contains all three adjective-pair families. The production model comparison reported in Section 5 uses all 9,100 canonical annotated observations over the 15 utterance categories and three adjective-family pairings. The size–colour marginal is retained as the theoretical anchor and descriptive subset.

Table A1

Participant exclusions and analysis-table sizes. Criterion counts overlap within study.

Experiment	Starting and final N	Participant exclusions	Analysis table used in manuscript
Study 1a slider ratings	150 $\rightarrow$ 134	8 with more than three missing trial-list entries; 8 with extreme completion time; 2 with more than three control-item failures; 6 with more than three “none fits” responses on experimental items; 16 unique participants excluded.	Broad cleaned critical table: 10,606 observations. Reported size–colour slider subset: 3,485 observations.
Study 1b slider replication	132 $\rightarrow$ 104	6 missing-list failures; 1 duration failure; 3 control failures; 24 experimental none-fit failures; 28 unique participants excluded.	Cleaned replication table: 8,379 critical observations. Exact design-equivalent tuples: 1,080.
Production study	149 $\rightarrow$ 113	8 with extreme completion time; 25 with more than 80% identical responses; 7 with more than 10% missing or unannotatable responses; 36 unique participants excluded.	Canonical full-data table: 9,100 annotated observations. All three adjective-family pairings enter the final production-model comparison.

The slider-rating materials used the fixed adjective vocabulary in Table A2. The raw stimulus table encodes colours and forms with compact feature labels; the interface displayed the corresponding German adjective forms in phrases such as *den großen blauen Aufkleber*.

Table A2

Adjective vocabulary in the slider-rating item descriptions.

Property class	Raw stimulus values	German adjective forms displayed in the item text
Size	Numeric size levels 1–6, 9, 10	<i>großen, kleinen</i>
Colour	blue, orange, grey, brown, green, black	<i>blauen, orangen, grauen, braunen, grünen, schwarzen</i>
Form	circle, diamond, heart, quadrat, star, triangle	<i>runden, diamantförmigen, herzförmigen, quadratischen, sternförmigen, dreieckigen</i>

The final modelling table is reconstructed deterministically from the raw stimulus records. Observed colour and form values are recoded into target–object relations without stochastic nuisance-feature generation; this recoding is separate from the lexical forms shown to participants.

Production responses were coded into the 15 ordered adjective categories listed in Table A3. The data files use D for the size or dimension adjective. The model exposition uses S; the two notations refer to the same adjective class. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size expressions, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table A3

Production coding dictionary. In the data files, D corresponds to the size adjective S ; C and F correspond to colour and form.

Length	Codes
One adjective	D, C, F
Two adjectives	DC, DF, CD, CF, FD, FC
Three adjectives	DCF, DFC, CDF, CFD, FDC, FCD

The behavioural hypotheses were evaluated through parallel frequentist mixed-effects tests and Bayesian predictive contrasts. Both analyses used the same fixed-effects design and random-effects structure. The later Bayesian production-model comparison addresses predictive differences among the proposed computational mechanisms over the complete response space. For the slider analysis, ratings were reoriented toward size-first, divided by 100, and centred by subtracting 0.5. The reported linear mixed-effects model included fixed effects of referential context, size discriminability, and their interaction, plus participant intercepts and context slopes:

$$y_{\text{slider}} \sim \text{context} \times \text{size discriminability} + (1 + \text{context} \mid \text{participant}).$$

Adding an item intercept yielded an estimated variance of zero and a singular fit. The reported participant-slope model converged without a singularity. In Study 1b, the context-by-discriminability interaction was small, $\chi^2(2) = 1.178$, $p = .555$. For production size-initiality, the logistic mixed-effects model used a binary outcome indicating whether the coded response began with the size adjective:

$$y_{\text{size-initial}} \sim \text{context} \times \text{size discriminability} + (1 + \text{context} \mid \text{participant}) + (1 \mid \text{item}).$$

For production over-informativeness, the descriptive model used the same fixed and random structure with a binary outcome indicating a three-adjective response in both-necessary

contexts or a multi-adjective response in one-property-sufficient contexts:

$$y_{\text{over-informative}} \sim \text{context} \times \text{size discriminability} + (1 + \text{context} \mid \text{participant}) + (1 \mid \text{item}).$$

The two production models converged without singular fits. For size-initiality, the planned discriminability contrast was supported in the both-necessary context ($p = .004$) and uncertain in the size- and colour-sufficient contexts ($p = .36$ and $p = .60$). For over-informativeness, the corresponding contrast was supported in the size-sufficient context ($p < .001$) and uncertain in the other two contexts ($p = .43$ and $p = .11$). The analysis script `paper/scripts/behavioural_robustness.R` writes the reported marginal estimates, planned contrasts, likelihood-ratio tests, and convergence checks to four CSV files under `paper/data/`.

The complementary Bayesian analyses retained these outcome definitions and random-effects structures. The slider models used a Gaussian likelihood, correlated participant intercepts and context slopes, $\text{Normal}(0, .30)$ priors on fixed effects, $\text{HalfNormal}(.20)$ priors on participant standard deviations, and a $\text{HalfNormal}(.25)$ prior on residual standard deviation. The production models used Bernoulli likelihoods, the same correlated participant effects, an item intercept, a $\text{Normal}(0, 2)$ intercept prior, $\text{Normal}(0, 1.5)$ priors on the remaining fixed effects, and $\text{HalfNormal}(1)$ priors on participant and item standard deviations. Participant-effect correlations received an $\text{LKJ}(2)$ prior. Four vectorized chains each provided 1,000 retained draws after 1,000 warm-up iterations. All four fits had zero divergences, no maximum-tree-depth hits, $\hat{R} \leq 1.01$, and minimum effective sample sizes above 900. For each posterior draw, posterior-predictive cell expectations were computed for a new participant and, in production, a new item by integrating over the fitted random-effects distributions. The planned predictive hypotheses were then evaluated on the rating or response-probability scale. The script `analysis/behavioural_bayes.py` writes the cell predictions, planned hypothesis contrasts, and inference diagnostics to `paper/data/behavioural_bayesian_predictions.csv`, `paper/data/behavioural_bayesian_hypotheses.csv`, and

`paper/data/behavioural_bayesian_diagnostics.csv.`

Appendix B

Final Production Model Specification

10.1 Deterministic Input Reconstruction

The final production input is reconstructed one-to-one from the raw stimulus table using participant list, condition, and item identifiers. Each retained response is paired with one observed 6×3 display tensor containing size, colour, and form values. An earlier preprocessing pipeline generated nuisance-feature values stochastically for some nominally irrelevant dimensions; the final analysis removes this transformation. Legacy first-draw, last-draw, and agreement-subset analyses are retained only as reproducibility sensitivities. The architecture and semantic-boundary conclusions are qualitatively stable across those legacy specifications.

10.2 Graded Size Meaning

Size is represented by a graded threshold over the scene-level comparison class $\mathcal{C}$ (Franke et al., 2019; Schmidt et al., 2009):

$$\theta_\delta(\mathcal{C}) = Q_{.80}(\mathcal{C}) - \delta \left(Q_{.80}(\mathcal{C}) - Q_{.20}(\mathcal{C}) \right), \quad (\text{B1})$$

where $Q_{.20}$ and $Q_{.80}$ are probability-weighted size quantiles and δ controls threshold selectivity. The implementation applies a perceptual-blur function around this threshold, yielding graded size-membership values. Equations (1), (2), (4) and (5) define how this meaning enters the literal listener and the three production accounts.

10.3 Component Inventory

Table B1

Final production-model components.

Component	Status	Role
Observed display input	Fixed	All visible size, colour, and form values
Size threshold and blur	Fixed constants	Graded size meaning in the scene comparison class
Colour reliability	Fixed	Soft determinate colour meaning
Form reliability	Fixed neutral	Form production supported by the task link
Pragmatic rationality	Estimated, hierarchical	Participant variation in response concentration
Stable-order score Ω	Fixed feature	Relative order expectation within adjective sets
Stable-order weight β_i	Estimated, hierarchical	Participant variation in stable order expectation
Salience coefficient	Estimated	Display prominence in candidate utility
Task-link coefficients	Estimated	One-word support, form backup, length and reliability terms
Successive-choice weight κ	Estimated in plan-guided production	Shared production parameter

Insert the frozen prior inventory for the final plan-guided production model.

Appendix C

Production Account Comparison and Diagnostics

Table C1

Final architecture contrasts after participant-specific stable-order variation is controlled.

Contrast	Δ ELPD	Paired SE	Participant-clustered SE	Bootstrap 95% interval
Plan-guided minus global	229.605	20.900	81.097	[58.765, 376.160]
Plan-guided minus fully in- cremental	422.387	35.615	108.714	[221.275, 647.382]
Fully incremental minus global	-192.782	—	169.152	[-553.149, 108.336]

All three final models used four chains, 1,000 warm-up iterations, and 1,000 retained draws. They had zero divergences, maximum $\hat{R} \leq 1.006$, minimum BFMI above .64, and no observation with Pareto $k > .7$. The exact endpoint audit returned maximum probability and log-probability differences of zero when plan-guided production was evaluated at $\kappa = 0$ and $\kappa = 1$.

Appendix D

Model-Development Checks

Form package. The selected form specification fixes form reliability at .50 and estimates a form-use coefficient in the task link. Two fixed informative alternatives tied form reliability to the colour value, with either the original or a strongly shrunk backup prior. The alternatives had lower predictive accuracy in the full response space. This comparison supports the selected form specification as an auxiliary predictive component and leaves participants' lexical semantics for form open.

Stable order source. The selected specification uses the German language-model residual alone. Adding a reliability-order term yielded worse conditional-order prediction within fixed adjective sets. Models that removed the language-model residual yielded invalid posterior or PSIS evidence. The comparison rejects the proposed reliability addition and leaves the cognitive source of stable order expectations open.

Task link. The reduced task-link comparison removed the sufficient-one-word bonus, form-use term, three-word penalty, and size-reliability terms as one bundle. Plan-guided production retained its relative advantage over both endpoints, while all reduced models lost substantial absolute predictive accuracy.

Insert the compact numerical tables for the auxiliary form, stable-order, and reduced task-link comparisons.

Appendix E

Participant Hierarchy Audit

The participant hierarchy audit compared the shared planning parameter with participant-specific planning, participant-specific length bias, and participant-specific stable-order weighting. Participant-specific planning gained 317.301 ELPD units over the shared model and 317.891 over participant-specific length bias. It differed from participant-specific stable-order weighting by -2.184 ELPD units, with participant-bootstrap interval $[-154.051, 137.077]$. The participant-specific planning model also had maximum $\hat{R} = 1.0133$, two observations with Pareto $k > .7$, and a failed participant-scale dependence screen. The audit therefore leaves participant-specific planning and stable-order heterogeneity unidentified. Descriptive correlations between fitted participant parameters and response summaries decode the latent parameter within that fit and provide no independent validation.

Appendix F

Endpoint Identity and Architecture Audit

At common parameter values, plan-guided production with $\kappa = 0$ exactly recovers global production, and with $\kappa = 1$ exactly recovers fully incremental production. The participant-specific order mapping, terminal response-policy kernel, and response support are identical across the three architectures. Maximum category-probability and log-probability differences were zero at a numerical tolerance of 10^{-10} .

Appendix G

Sequential Context Updating for Size

For the hierarchical semantic path, let H_{j-1} be the state after the syntactically inner modifiers have composed. The context-updating listener recomputes the size threshold from this hierarchically prior state:

$$\llbracket S \rrbracket_{\text{upd}, H_{j-1}}(o) = \llbracket S \rrbracket_{\theta_\delta(H_{j-1})}(o), \quad (\text{G1})$$

The context-fixed listener anchors the threshold to the scene prior:

$$\llbracket S \rrbracket_{\text{fix}}(o) = \llbracket S \rrbracket_{\theta_\delta(P_0)}(o). \quad (\text{G2})$$

For *big blue*, hierarchical composition applies the colour restriction before interpreting size. The distinct surface production calculation follows left-to-right order and uses the validated factorisation in which every meaning is applied once. The matched observed-display comparison yielded no practically meaningful predictive difference between fixed and sequentially updated comparison classes.

Appendix H

Simulation Parameter Sweeps

Figures H1 to H3 show how the three fixed semantic parameters affect communicative success across the two speaker architectures.

Table H1

Simulation parameter grid

Parameter	Grid values	# levels
Speaker architecture	Incremental, global	2
Semantic regime	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
ν	0.90, 0.92, 0.94, 0.96, 0.98	5
k	0.10, 0.30, 0.50, 0.70, 0.90	5
w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Random context samples per combination	1,000	
Total trials	12,000,000	

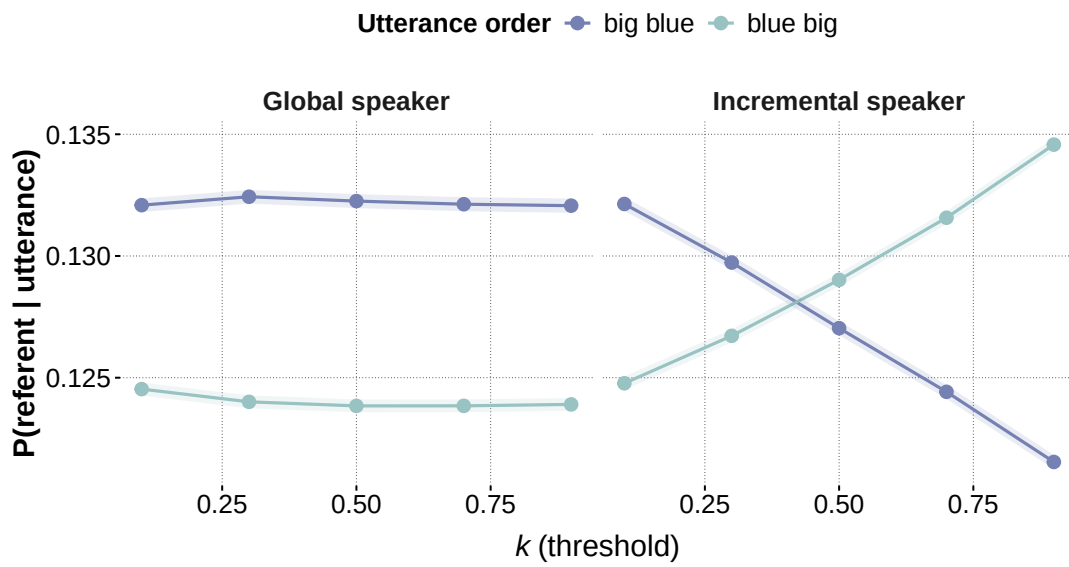


Figure H1

Effect of the threshold parameter k on communicative success in the simulation.

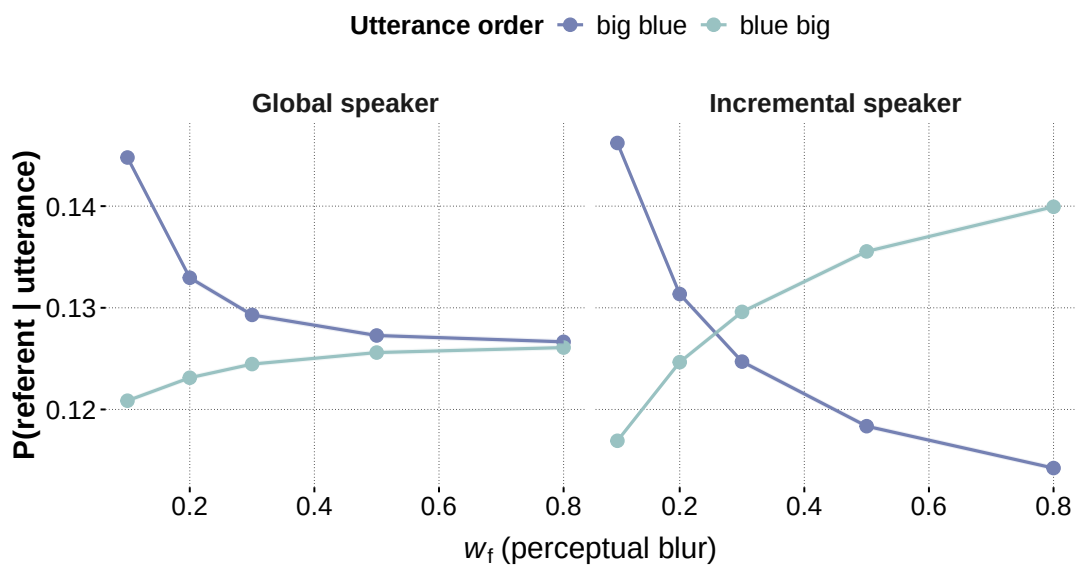


Figure H2

Effect of perceptual blur w_f on communicative success in the simulation.

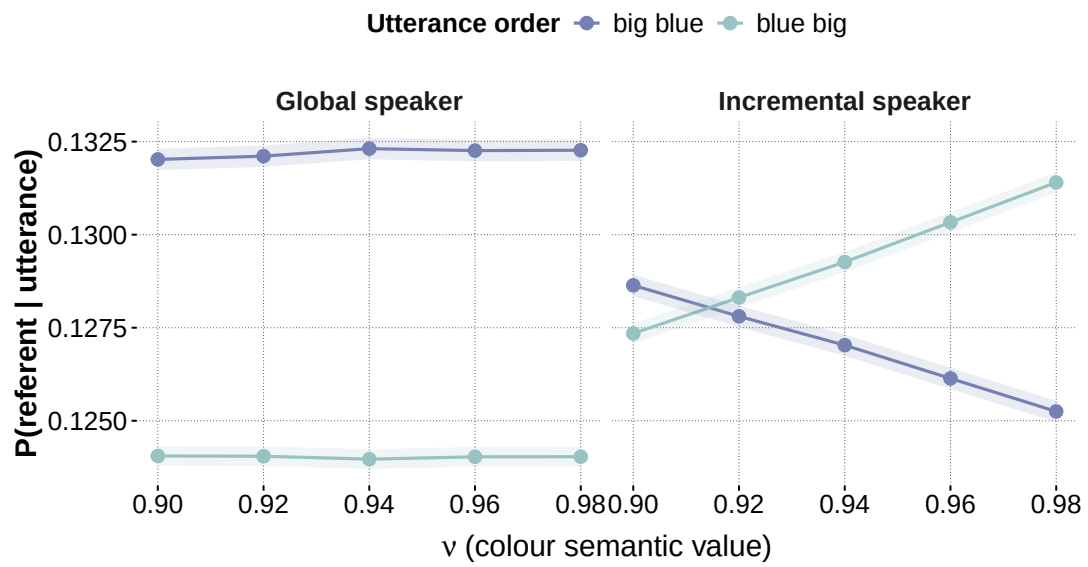


Figure H3

Effect of the colour semantic value ν on communicative success in the simulation.

Appendix I

Slider Colour-Reliability Audit

The slider-rating model fixes colour reliability ν at 0.80 in its diagnostic specification. Production uses the softer colour value $\nu_{\text{colour}} = 0.59$ in the production speaker (Section 5); for parity, we apply the same kind of sensitivity audit to the slider’s incremental speakers. The audit samples $\nu \sim \text{Uniform}(0.5, 0.999)$ within the model and recomputes the literal listeners at each MCMC step. Every other component of the incremental speaker remains unchanged. Table I1 reports the resulting posteriors alongside the originally fixed- ν fits.

Table I1

Slider colour-reliability audit. Fixed- ν rows use $\nu = 0.80$; free- ν rows sample $\nu \sim \text{Uniform}(0.5, 0.999)$ and recompute literal listeners each step. “ELPD” is PSIS-LOO; ELPD loss and dse are relative to the incremental, context-updating fixed- ν model. All four models use 4 chains $\times$ 500 warmup $\times$ 500 samples; maximum split- $\hat{R}$ rounds to 1.01 for the free- ν models.

Model	ELPD	ELPD loss	dse	α (sd)	b (sd)	ν (94% HDI)
Incremental, updating, fixed $\nu = 0.80$	-2,774.1	0.0	—	0.69 (0.06)	0.58 (0.06)	—
Incremental, fixed, fixed $\nu = 0.80$	-2,774.1	0.06	0.18	0.69 (0.06)	0.58 (0.06)	—
Incremental, updating, free ν	-2,775.0	0.89	1.08	1.53 (0.61)	0.41 (0.15)	0.63 [0.50, 0.82]
Incremental, fixed, free ν	-2,774.3	0.23	1.00	1.48 (0.55)	0.42 (0.13)	0.63 [0.51, 0.79]

Three points stand out. First, ν is weakly identified in the slider data: the updating credible interval includes the fixed value 0.80, the fixed-semantics credible interval reaches 0.79, and both posteriors place substantial mass near the lower prior boundary at 0.5. The production estimate has a narrower credible interval, 0.59 [0.56, 0.62]. Second, the nominal PSIS-LOO losses are all smaller than their dse (loss/dse < 1 throughout), and the shared

PSIS warnings limit these values to diagnostic use. Third, under the freed posterior the rationality parameter α moves to roughly 1.5 (from 0.69) and the bias intercept b moves below 0.5 (from 0.58), because ν and the other parameters trade off: a softer colour predicate forces a stronger rationality term and reduces the residual size-first bias absorbed by b . We therefore retain the fixed $\nu = 0.80$ for the slider model in the main text and treat α and b at that specification as the reported quantities. The fixed- ν estimate $b = 0.58$ remains conditional on this specification, and its archived credible interval includes 0.5. The broader conclusion is unchanged: the slider response space weakly identifies semantic scale parameters, and the fitted PSIS-LOO comparisons remain diagnostic.